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**B.TECH.-Fourth SEMESTER
END-SEMESTER THEORY EXAMINATION,
April-May 2026**

Course Code-CMMTC13

Course Title-Applied Linear Algebra

Time: 3:00 hours

Max Marks-50

Note: Attempt all the five questions. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q.1.	Attempt any 2 parts of the following		
1(a)	Let $V = \{(x, y) \in \mathbb{R}^2 : x, y > 0\}$ with operations defined by $(x_1, y_1) \oplus (x_2, y_2) = (x_1 x_2, y_1 y_2)$ and $a \odot (x, y) = (x^a, y^a)$, $a \in \mathbb{R}$. Show that $(V, \oplus, \odot)$ forms a vector space over $\mathbb{R}$.	5	CO1
1(b)	Check whether the set $S = \{(x+1)^2, x(x+1), x^2+3\}$ is a basis of the space $P_2(\mathbb{R})$ or not.	5	CO1
1(c)	Prove that the sets $A = \{(x, y, z) \in \mathbb{R}^3 : x - y + 2z = 0\}$ and $B = \{t(1, -1, 2) : t \in \mathbb{R}\}$ are subspaces of $\mathbb{R}^3$. Determine whether $\mathbb{R}^3$ is the direct sum of A and B or not.	5	CO2
Q.2.	Attempt any 2 parts of the following		
2(a)	Verify rank-nullity theorem for the matrix $A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 3 & 4 & -1 & 2 \\ -1 & -2 & 5 & 4 \end{bmatrix}$	5	CO2
2(b)	Let $T : P_3(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ be a linear transformation such that $T(f(x)) = f''(x) - 4f'(x) + f(x)$. Find the associated matrix $[T]_{\beta}$ for the basis $\beta = \{x, 1+x, x+x^2, x^3\}$.	5	CO2
2(c)	Show that the linear transformations $T, S \in \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(x, y) = (x - y, x + y)$ and $S(x, y) = (2x, 3y)$ are invertible. Also, find $(T \circ S)^{-1}$.	5	CO2
Q.3.	Attempt any 2 parts of the following		
3(a)	Find the Jordan canonical form of the matrix $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 4 & -6 & 4 \end{bmatrix}$	5	CO3

3(b)	Diagonalize the matrix $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ and also evaluate the matrix norms $\ A\ _1$, $\ A\ _2$ and $\ A\ _\infty$.	5	CO3
3(c)	Verify the Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$ and hence compute A^{-1} .	5	CO3
Q.4.	Attempt any 2 parts of the following		
4(a)	Find an orthonormal basis for the space $\mathbb{R}^3$ by applying the Gram-Schmidt process to the vectors $(1, 0, 1)$, $(1, 0, -1)$ and $(0, 3, 4)$.	5	CO4
4(b)	Define unitary and normal operators. Verify that the matrix $A = \begin{bmatrix} 2 & i \\ i & 2 \end{bmatrix}$ is normal. Diagonalize A by a unitary matrix U .	5	CO4
4(c)	Write properties of an inner product space V and prove that $\ x + y\ ^2 + \ x - y\ ^2 = 2\ x\ ^2 + 2\ y\ ^2$, where $x, y \in V$.	5	CO4
Q.5.	Attempt any 2 parts of the following		
5(a)	Verify that the function $H : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by $H((x_1, x_2, x_3), (y_1, y_2, y_3)) = x_1y_1 + 3x_2y_1 - x_1y_3 + 2x_3y_2$ is a bilinear form. Also, find its matrix representation with respect to the basis $\beta = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$.	5	CO5
5(b)	Find the rank, signature and index for the matrices representing each of the following quadratic forms: (i) $x^2 + 6xy + 2y^2$, (ii) $x^2 - 2y^2 - 3z^2 + 4xy + 6xz - 8yz$.	5	CO5
5(c)	Let $H : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined by $H(x, y) = x^T A y$, where $A = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 0 & 4 \\ -1 & -4 & 5 \end{bmatrix}$. Decompose H into the sum of a symmetric and a skew-symmetric bilinear forms. Also, find the quadratic form associated with the symmetric part.	5	CO5

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